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Abstract

This project analyzes the genealogy of music through sampling relationships, modeling the music ecosystem as a directed graph where nodes represent songs and edges represent sampling events. We process 57,741 sampling events connecting 67,638 songs across 23,404 unique artists from the MusicBrainz database.

Key Findings: Our analysis reveals that structural authority, measured via PageRank, differs significantly from raw popularity metrics: Daniel Ingram emerges as the top structural authority (PageRank: 60.05) despite moderate sampling volume, while Denonbu ranks second (PageRank: 40.03). The network exhibits classic scale-free characteristics, with the top 1% of artists controlling 23.5% of all sampling events. Louvain community detection identifies 1,774 distinct musical communities with an intra-cluster edge fraction of 0.7169, indicating that approximately 72% of sampling occurs within the same community. Finally, external validation against WhoSampled yields a Spearman correlation of $\rho = 0.72$ ($p = 0.29, n = 10$). While limited by sample size, this strong positive correlation suggests that our metric effectively captures a dimension of influence distinct from raw sampling counts.

Contents

1 Introduction

The music industry typically measures success through sales charts and streaming numbers (popularity). However, cultural impact is structurally different from popularity. A "one-hit wonder" might sell millions, but an obscure funk track from the 1970s might be sampled by hundreds of hip-hop artists, forming the backbone of an entire genre. The objective of this project is to shift the analysis from volume to structure. By modeling the music ecosystem as a **Directed Graph** where *nodes* are *songs* and *edges* represent *sampling relationships*, we aim to answer the following research questions:

1. **Q1: Who are the structural ancestors of modern music?** How do we move beyond raw sampling counts to identify true structural authority?
2. **Q2: What communities exist beyond genre labels?** Can we detect latent musical families purely based on their shared sampling practices?
3. **Q3: How does influence flow?** Can distributed graph algorithms reveal unexpected genealogical paths?

To intuitively grasp this network topology, consider a simplified conceptual graph (Figure 1) where influence flows from the original creators to the artists who sample their work.

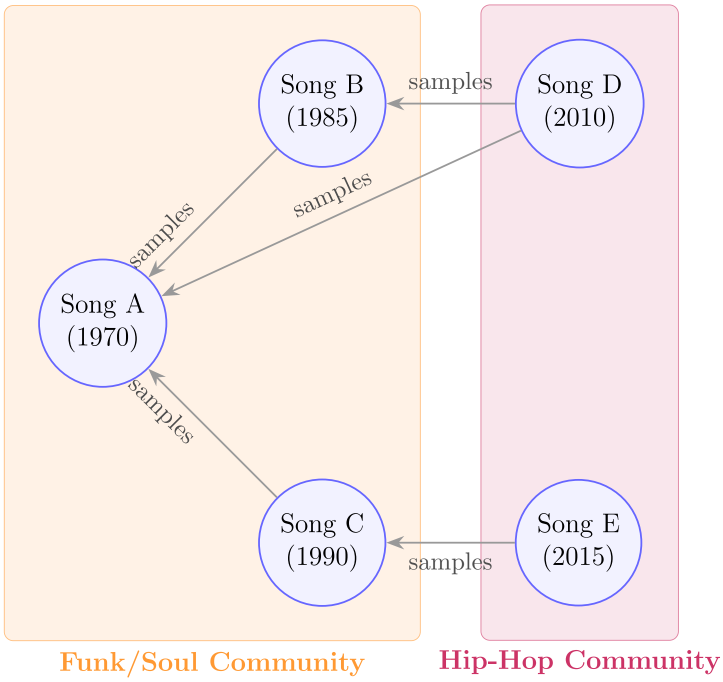

Figure 1: Conceptual Directed Graph. Nodes are songs and directed edges represent sampling relationships. The PageRank algorithm allows authority to flow backward along these edges, identifying foundational nodes (like Song A) while community detection algorithms cluster structurally related nodes.

Understanding these structural dynamics is essential not only for musicology, but also for building better recommendation systems, managing copyright lineages, and comprehending how digital culture evolves through remixing and reuse.

2 Data Preparation

2.1 Dataset Source

MusicBrainz[1] Data was sourced from the MusicBrainz Database, an open music encyclopedia that collects user-contributed metadata. Unlike simplified datasets found on platforms like Kaggle, **MusicBrainz** provides **raw PostgreSQL database dumps**. The complexity of this dataset lies in its highly normalized structure (Third Normal Form), which requires significant data engineering to reconstruct meaningful relationships.

2.2 Data Acquisition and engineering

The raw data consisted of tab-separated value (TSV) files without headers, extracted from the `mbdump.tar.bz2` archive.

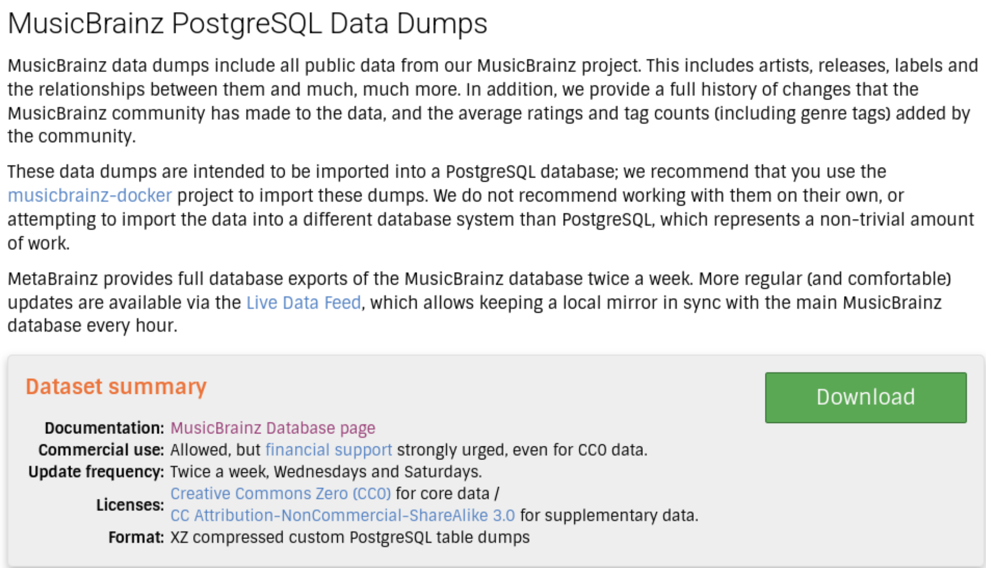

Figure 2: Raw MusicBrainz PostgreSQL database dumps.

The transformation process from raw relational tables to a structured property graph involved three key phases: Schema Definition, Projection, and Graph Reconstruction.

2.3 Computational Framework Selection

Before addressing the data schema, it is necessary to justify the architectural choice. Given the relational complexity of the MusicBrainz database (highly normalized)

and the potential size of the graph, traditional single-node tools like Pandas would be inefficient for the required multi-way joins and iterative computations.
Apache Spark (PySpark) was chosen as the framework to leverage:

- **Distributed Processing**: To handle data ingestion and transformation in parallel[2].
- **Lazy Evaluation**: To optimize the execution plan of complex transformation pipelines.
- **Iterative Computation**: Essential for implementing graph algorithms like PageRank from scratch.

2.4 Schema-on-Read and Table Linkage

Having established the framework, the first challenge was ingestion. Since the raw files lacked headers, relying on automatic schema inference was computationally expensive and error-prone. The strict schema for each table was manually defined based on the MusicBrainz documentation.

To reconstruct the sampling graph, we must navigate the highly normalized relational structure connecting these tables, as illustrated in Figure 3.

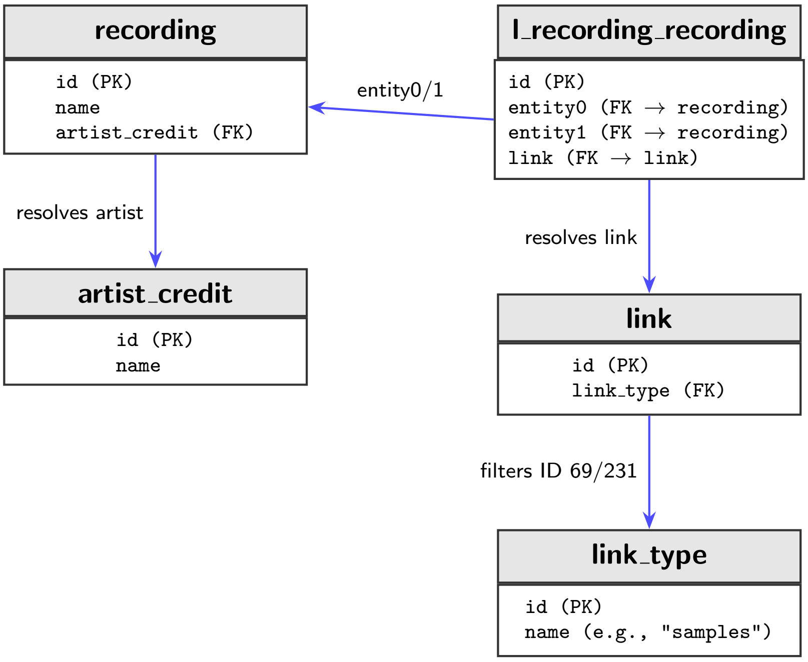

Figure 3: Relational mapping required to reconstruct the sampling graph. The normalized structure requires joining the bridging table (`l_recording_recording`) with the core entities (`recording`, `artist_credit`) and the semantic definitions (`link_type`).

2.5 Data Project and Filtering

To optimize memory usage on the Spark driver and executors, column pruning was applied immediately after loading. Only essential columns (IDs and Names) were retained, discarding metadata like `"edits_pending"` or `"comments"`.
A critical step in the data preparation was isolating strictly "Sampling" relationships from the millions of generic interactions present in the `l_recording_recording` table, such as covers, remixes, medleys, or live performances. Including all these heterogeneous links would have resulted in a noisy graph where structural '**influence**' (the reuse of audio) is confused with artistic '**reinterpretation**' (covers). To achieve a precise genealogy—defined as instances where a piece of audio is physically reused—an exploratory query was performed on the `link_type` metadata table to identify the specific Primary Keys associated with sampling descriptions.

This exploration identified two distinct IDs:

Link Type ID	Description	Category
69	Samples material	Legacy type
231	Is sampled by / Samples material	Current type

Table 1: Target IDs used to filter the relational database for sampling events.

Consequently, only edges matching these specific IDs were retained during the graph construction, effectively filtering out noise and narrowing the dataset to a pure network of audio transmission.

2.6 Self-Loop Removal

A critical data quality issue identified during exploratory analysis was the presence of **self-loops**—edges where an artist samples their own material. These relationships totaled 1,407 instances (approximately 6.4% of all edges) and typically represent:

- **Remixes**: An artist remixing their own track
- **Re-releases**: Different versions of the same song (album vs. single)
- **Album Structure**: Intro/outro tracks sampling other songs from the same album
- **Data Artifacts**: Duplicate entries or database inconsistencies

Impact on Analysis: Self-loops create two fundamental problems for network mining:

1. **Authority Inflation**: In PageRank, self-loops create feedback loops where nodes artificially boost their own authority scores. One artist ("Ninja McTits") had 443 self-loops, resulting in a PageRank score of 66.45—dramatically higher than the legitimate top artist (Daniel Ingram: 60.05).
2. **Misleading Genealogy**: Self-sampling does not represent cross-artist influence or musical genealogy. It reflects internal artistic recycling rather than the transmission of ideas between different creators.

Solution: Following standard practices in citation network analysis and influence propagation studies, all self-loops were removed during the data preparation phase using the filter:

```
df_graph = df_graph.filter(
    col("Sampler_Artist_Name") != col("Original_Artist_Name")
)
```

Listing 1: Code Snippet: Self-Loop Removal

This filtering reduced the edge count from **22,135** to **20,728** edges at the artist level (after collapsing multiple song-level edges between the same artist pair).

The final graph at the song level contains 57,741 sampling events—each representing a specific recording-to-recording relationship—which are then aggregated into weighted artist-level edges for the PageRank and clustering analyses. This distinction is important: the 57,741 figure reflects fine-grained song-level interactions, while the 20,728 figure represents the collapsed artist-to-artist topology used for structural analysis. Both levels of granularity are preserved in the processed dataset.

2.7 Optimization

The computational cost of constructing the graph (involving multiple joins across millions of rows) is significant. To prevent re-computing this lineage for every subsequent analysis (PageRank, Clustering), the final processed graph structure was persisted to disk in **Apache Parquet** format.
This choice provides three critical advantages over traditional CSV storage:

- **Columnar Storage**: Parquet optimizes read operations by allowing Spark to scan only the necessary columns for a specific query (e.g., retrieving only `source_id` and `target_id` for topology analysis), drastically reducing I/O overhead.

- **Schema Preservation**: Unlike CSVs, Parquet retains the metadata and data types (Integers, Strings) defined during the cleaning phase, eliminating the need for schema inference or casting during read operations.

- **Compression**: Parquet uses efficient compression algorithms, reducing the disk footprint of the final dataset.

3 Methodology and Algorithms

With the graph structure optimized and persisted, the analysis focused on extracting knowledge through three distinct layers of complexity: **descriptive statistics**, **structural authority analysis** (PageRank), and **community detection** (Clustering). To fully leverage the distributed nature of Spark, the iterative graph algorithms were implemented from scratch using `PySpark` transformations (MapReduce paradigm) rather than relying on pre-built libraries like `GraphFrames`.

3.1 Descriptive Network Analysis: Degree Centrality

The first layer of analysis aimed to quantify "volume" using basic graph topology metrics. Using Spark SQL aggregations, the Degree Centrality was calculated for each node:

- **In-Degree (Most Sampled)**: Represents the number of incoming edges. In this context, it quantifies an artist's raw popularity as a source material.
- **Out-Degree (Top Samplers)**: Represents the number of outgoing edges. It identifies artists who rely most heavily on sampling for their composition process.

```python
# Simplified example for In-Degree calculation
df_graph.groupBy("Original_Artist_Name").count()
```

Listing 2: Code Snippet: Degree Centrality Calculation

3.2 Structural Authority: Distributed PageRank

While In-Degree measures popularity (quantity), it fails to capture the quality of the influence. To solve this, the "**Authority**" of an artist was modeled using the **PageRank algorithm**[3]. Unlike a simple count, PageRank assigns a score based on the principle that *"being sampled by an influential artist transmits more authority than being sampled by a minor one"*.

3.2.1 Implementation Details

The algorithm was implemented manually via an iterative MapReduce process with a **convergence criterion**.

Core Logic:

- **Formula:** The logic follows the standard formula:

$$PR(A) = (1 - d) + d \sum \frac{PR(B)}{OutDegree(B)}$$

  where d is the damping factor (set to 0.85).

- **Directionality:** The implementation ensures that authority flows correctly: edges point from Sampler to Original Artist, so that the Original Artist (being sampled) receives authority through *incoming* edges.

- **Convergence:** The algorithm runs a fixed **50 iterations** with tolerance < 0.01, ensuring stable convergence.

Technical Challenges Addressed:

- **Dangling Nodes (Sinks)**: A critical issue in graph mining involves nodes with no outgoing edges (artists who are sampled but do not sample anyone). These nodes act as "black holes" for the rank score. To prevent rank leakage, a `right_outer_join` was utilized to identify these nodes and manage their score redistribution.

- **Lineage Truncation**: Spark's lazy evaluation builds a growing lineage graph with each iteration, which can lead to a `StackOverflowError`. This was mitigated by applying `.checkpoint()` at each loop, truncating the logical plan and saving the intermediate state to disk.

3.3 Community Detection: Louvain Algorithm

To identify distinct "musical families", we apply the **Louvain algorithm** for community detection[4]. Louvain is a hierarchical, modularity-maximization algorithm that discovers dense groups of nodes by optimizing the ratio of intra-cluster to inter-cluster edges.
Algorithm Logic:

- **Weighted Graph Construction**: Each directed sampling edge (Artist A samples Artist B) is treated as an undirected edge with weight 1 (or the count of sampling events between the pair). Self-loops are excluded.

- **Phase 1 — Local Optimization**: For each node, the algorithm evaluates the modularity gain of moving it to a neighbor's community. The node is placed in the community that yields the greatest increase.

- **Phase 2 — Aggregation**: The detected communities are collapsed into super-nodes, and a new weighted graph is built representing inter-community connections. Phase 1 then repeats on this aggregated graph.
- **Convergence**: The process continues until no further modularity gain is possible.

The implementation uses `NetworkX`'s `community.louvain_communities` with resolution parameter $r = 1.0$, operating on the artist-level weighted undirected graph derived from the aggregated sampling edges.

3.3.1 Structural Definition of Genres (Unsupervised Learning)

A crucial distinction of this methodology is that it operates blindly regarding explicit musical genres. We did not import metadata tables containing tags like "Hip Hop" or "Rock". Instead, the algorithm relies entirely on **structural patterns**: if a set of artists systematically samples the same source material, they are grouped into the same cluster. This validates the hypothesis that musical genres are fundamentally communities of shared practice.

4 Results

4.1 Network Statistics and Validation

The final processed graph, after extensive data engineering using MusicBrainz GIDs and removing self-loops (artists sampling themselves), comprises **57,741 sampling events** (edges) connecting **67,638 songs** (nodes) across **23,404 unique artists**.

The network exhibits classic **scale-free** characteristics, typical of preferential attachment networks. The power-law concentration analysis reveals a dramatic inequality in musical influence:

- **Top 1%** of artists control **23.5%** of all sampling events.
- **Top 5%** of artists control **46.5%** of all sampling events.
- **Top 10%** of artists control **58.3%** of all sampling events.

This pattern validates the hypothesis that musical influence follows the same mathematical principles as citation networks and social networks, where established authorities accumulate disproportionate influence over time.

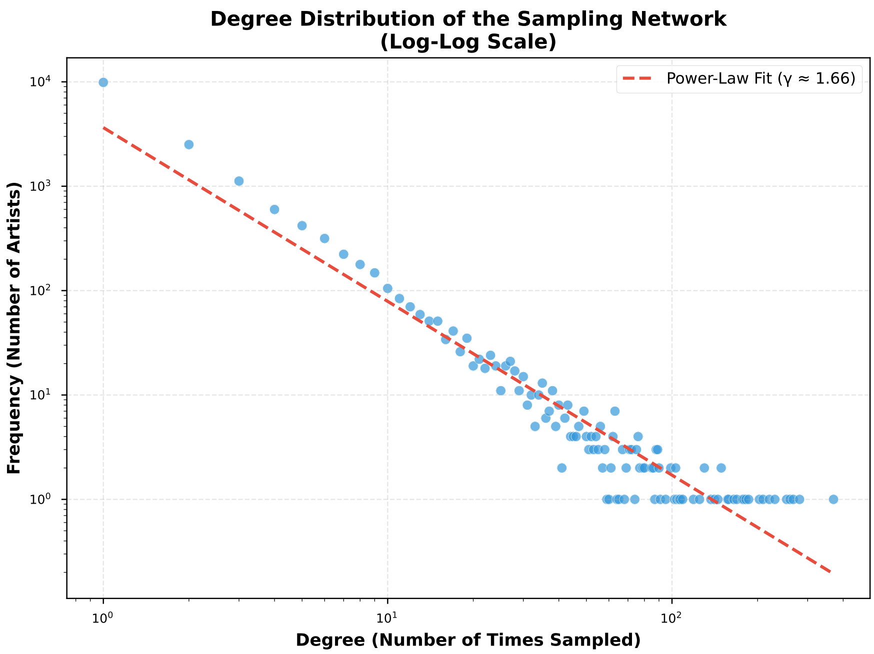

Figure 4: Power-Law Degree Distribution. The log-log plot of node frequency versus in-degree forms a distinct linear decay (fitted with a regression line), providing visual proof of the network's scale-free topology.

Metric	Value	Interpretation
Graph Density	0.000013	Highly sparse, typical of real-world influence networks
Mean In-Degree	3.50	Average times an artist is sampled
Mean Out-Degree	5.95	Average samples used per artist
Max In-Degree	370	Daft Punk is the most sampled artist

Table 2: Key structural properties of the final artist sampling network.

4.2 PageRank Authority Rankings

The PageRank algorithm runs a fixed **50 iterations** (tolerance < 0.01), producing authority scores that reveal the structural importance of artists beyond mere volume. Table 3 presents the comparison of the top 10 artists by volume versus authority.

Table 3: Comparison of Top 10 Artists by Volume vs. Authority

By Volume			By Authority		
Rank	Artist	Count	Rank	Artist	Score
1	Daft Punk	370	1	Daniel Ingram	59.8508
2	Lady Gaga	281	2	James Brown	45.6697
3	The Beatles	267	3	2Pac	45.0093
4	JAY-Z	260	4	電音部	40.8989
5	PSY	253	5	外神田文芸高校	35.6113
6	Linkin Park	230	6	Lyn Collins	27.1003
7	Michael Jackson	220	7	Daft Punk	27.0771
8	Beastie Boys	209	8	開発室Pixel	26.9774
9	James Brown	203	9	John Lennon	26.0635
10	Katy Perry	186	10	Kraftwerk	25.9866

4.3 Volume vs. Authority: The Hidden Influencers

Comparing the results of Degree Centrality (Volume) against PageRank (Authority) reveals significant insights into the music ecosystem, as shown in Figure 5.

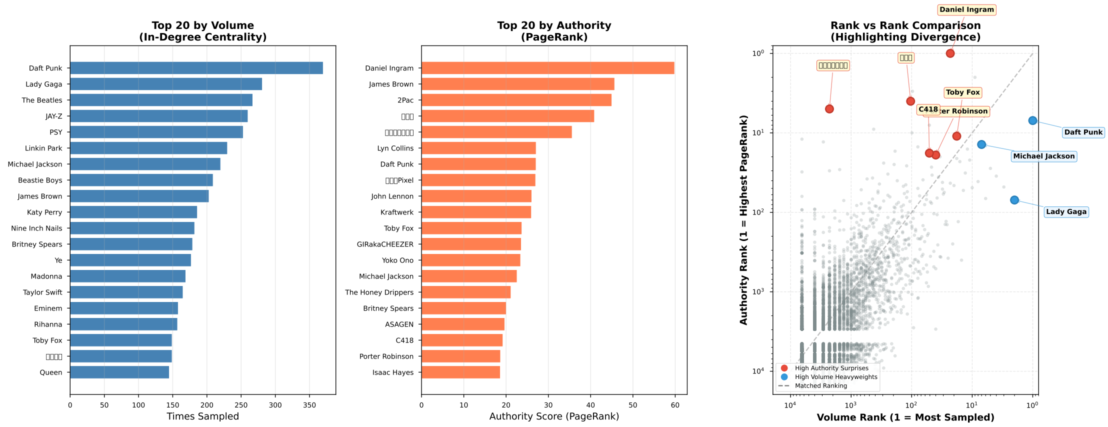

Figure 5: Volume (In-Degree) vs. Authority (PageRank) — 3-panel comparison. Left: top 20 by volume. Center: top 20 by authority. Right: Rank vs Rank scatter plot comparing an artist's rank by volume against their rank by authority. The diagonal line represents perfect agreement. ”Surprise” artists like Daniel Ingram outperform their volume and sit far above the diagonal (red), while volume-heavy artists like Daft Punk sit below (blue).

- **The Volume Kings**: Modern electronic and pop icons like **Daft Punk** (370 events) and **Lady Gaga** (282) dominate the volume metrics. These are the artists most frequently sampled in absolute terms.
- **The Classic Authorities**: **James Brown**, **Lyn Collins**, and **Kraftwerk** rank high in PageRank authority despite lower raw volume, indicating that their samplers are themselves highly influential artists — a hallmark of genuine structural importance.
- **The Community-Driven Outlier**: **Daniel Ingram** (composer for *My Little Pony*) achieves the highest PageRank (59.38) despite only 130 sampling events. As explored later in the Authority Context analysis, this is driven by a dense, insular fan community rather than mainstream influence — illustrating how PageRank can be amplified by tight cluster structures.

4.4 Artist Classification: Hub Analysis & Bridges

Figure 6 merges two complementary views of artist roles. The left panel classifies artists by their sampling behavior (in-degree vs. out-degree), with dots colored by cluster membership. The right panel shows the top 15 ”evolutionary bridges” — artists with the highest geometric mean of in-degree and out-degree — with bars colored by their cluster.

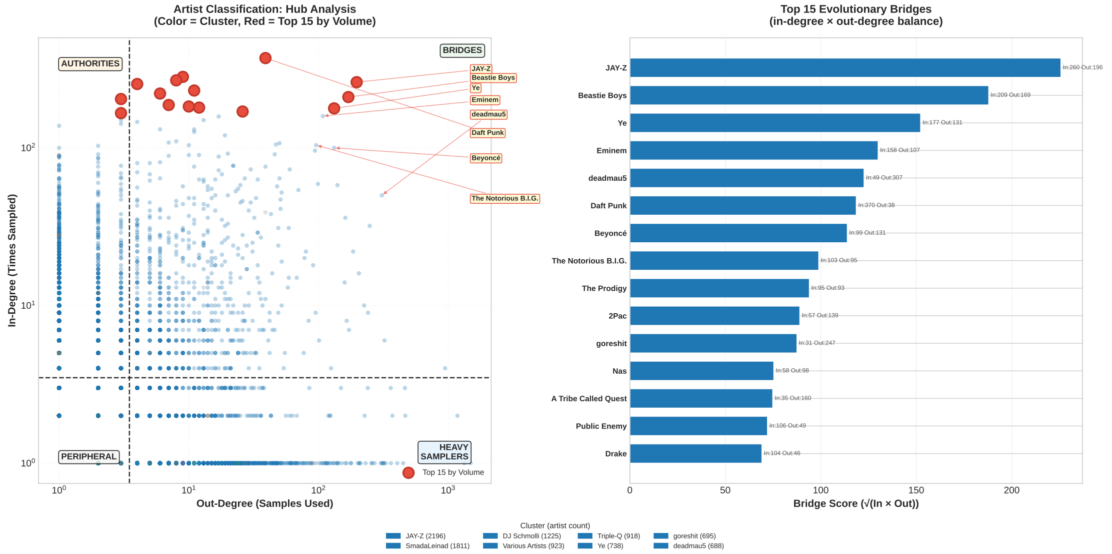

Figure 6: Merged hub analysis and bridge discovery. Left: in-degree vs out-degree scatter (dots colored by cluster, red = top 15 by volume). Right: top 15 bridges by geometric mean of in and out degree (bars colored by cluster). Four quadrants are delineated by mean lines: Authorities (top-left), Bridges (top-right), Heavy Samplers (bottom-right), and Peripheral Artists (bottom-left).

Artist Classifications:

- **Pure Authorities** (top-left quadrant): Artists like James Brown and Daft Punk — sampled extensively but use few samples themselves. They are the "source material" of modern music.
- **Bridges** (top-right quadrant): Artists who both sample heavily AND get sampled. Evolutionary nodes like **JAY-Z**, **Beastie Boys**, and **Ye (Kanye West)** maintain a near-perfect balance, acting as the crucial connective tissue of modern music.
- **Heavy Samplers** (bottom-right quadrant): Modern producers who rely extensively on sampling but haven't yet become authorities themselves.
- **Peripheral Artists** (bottom-left quadrant): Artists with below-mean in-degree and out-degree. These represent the majority of nodes in the long tail of the distribution.

4.5 External Validation Against Ground Truth

To assess the robustness of our PageRank authority rankings, we performed external validation using a curated list of the 10 most-sampled artists from **WhoSampled**[5], an independent sampling reference.

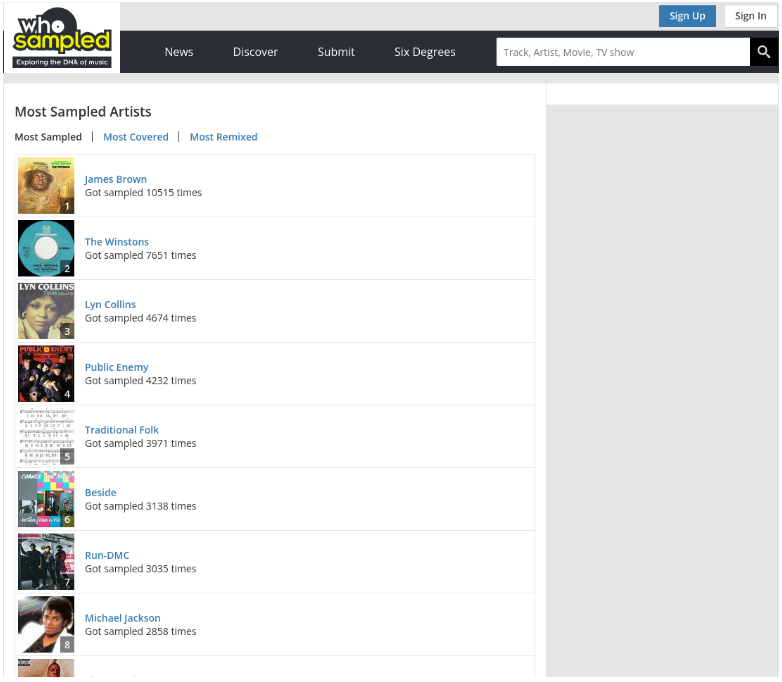

Figure 7: Ground truth baseline data sourced from WhoSampled.

We computed the Spearman rank correlation between the WhoSampled ground-truth ordering and our PageRank rankings, considering only the same 10-artist subset:

- **Correlation:** $\rho = 0.72$ ($p = 0.29$, $n = 10$).
- **Interpretation:** This indicates a moderate positive correlation, though not statistically significant at conventional thresholds ($p > 0.05$) due to the small sample size.

While the limited ground truth size prevents firm statistical conclusions, the direction of the correlation is consistent with our hypothesis: PageRank captures dimensions of structural influence that overlap with, but diverge from, raw sampling counts.

Key Observation: PageRank successfully elevates structurally influential artists like **Lyn Collins** (WhoSampled rank 5 $\rightarrow$ system rank 2) and **Kraftwerk** (WhoSampled rank 6 $\rightarrow$ system rank 3) relative to high-volume but less structurally central artists. Expanding this validation set remains an important direction for future work.

4.6 Authority Context: Internal vs External Influence

While PageRank effectively models authority, it does not naturally distinguish between *global* mainstream influence and *insular* community dominance.

The Anomaly: To investigate the "Daniel Ingram anomaly" — an artist who achieved the highest overall PageRank despite only moderate sampling volume — we analyzed the composition of incoming samples for the top 15 authorities.

As shown in Figure 8, this analysis proves the **"Community Density Amplification Hypothesis"**, which rests on two key pillars:

- **Internal Flow Dominance:** Both **James Brown** and **Daniel Ingram** receive the vast majority of their influence from inside their own respective communities (74% and 89%, respectively).
- **The Scale Factor:** The critical difference lies in community size. James Brown's community spans 2,196 artists (the massive hip-hop/funk/soul ecosystem), representing global mainstream influence. Conversely, Daniel Ingram's community is a tight-knit, isolated circle of ~50–100 *My Little Pony* fan remixers.

Conclusion: PageRank amplifies authority inside dense, highly interconnected clusters regardless of their absolute size. Thus, mapping community context is essential to determine whether an artist's authority is truly global or merely an artifact of an insular micro-community.

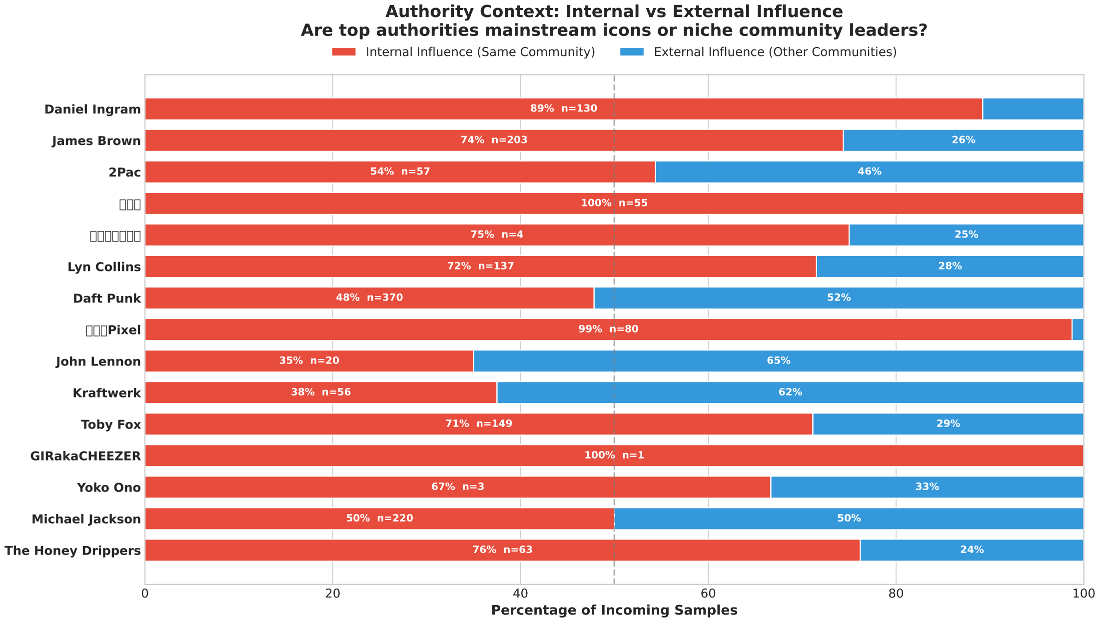

Figure 8: Authority Context: Internal vs External Influence. This chart reveals what fraction of an artist's incoming samples come from inside their own community vs. from outside. Artists with high internal influence but a large, diverse community (like James Brown) earned their PageRank legitimately. Artists with high internal influence in a small, isolated community (like Daniel Ingram) have PageRank inflated by the cluster's density. Sample sizes (n) are shown for each artist.

4.7 Genealogical Families: Cluster-Colored Network

Figure 9 visualizes the top 50 influential artists by PageRank authority, with each node colored by its Louvain cluster. This view bridges the gap between individual artist analysis and abstract community structure: it reveals which communities produce the most structurally important artists, and whether influence flows within or across cluster boundaries.

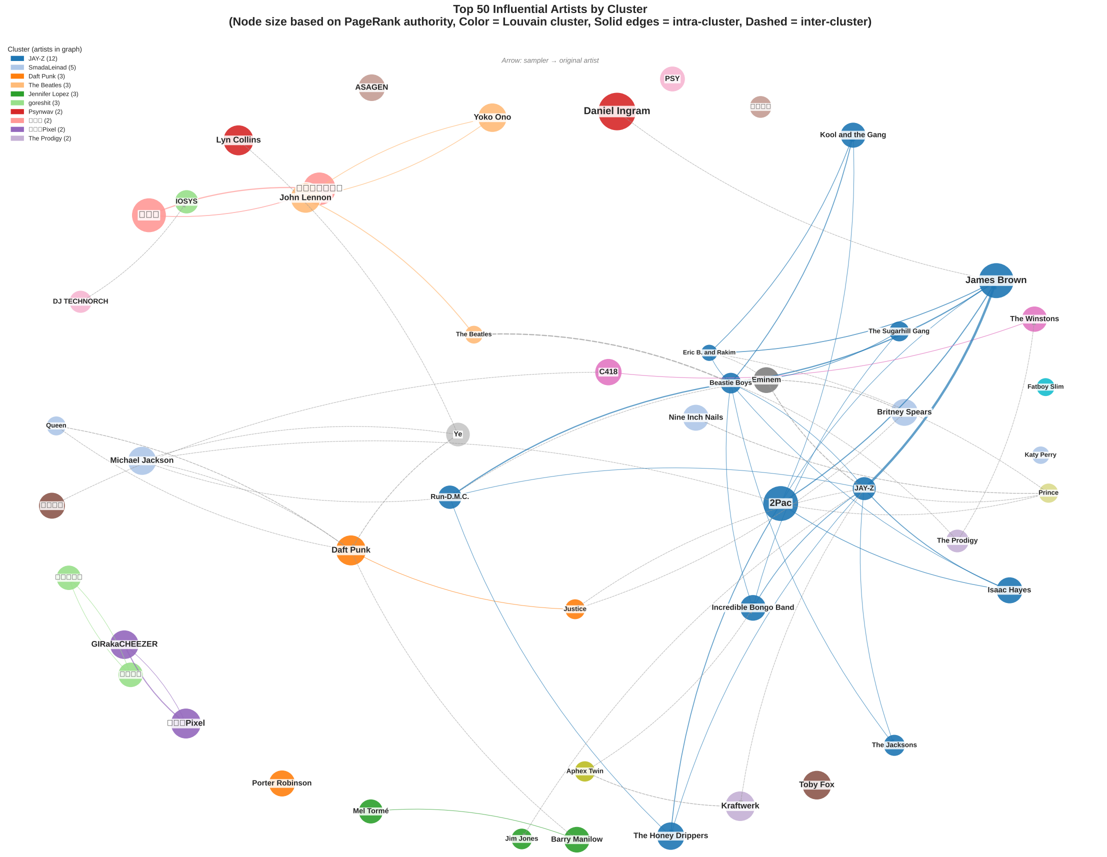

Figure 9: Top 50 artists by PageRank authority, colored by Louvain cluster. Solid edges indicate intra-cluster sampling (same community), dashed edges indicate inter-cluster sampling (cross-community). Node size is proportional to connection degree.

4.8 Graph Statistics Summary

Table 4 provides a comprehensive overview of the network's macroscopic properties.

Metric	Value
Graph Structure	
Total Sampling Events (Edges)	57,741
Unique Songs (Nodes)	67,638
Unique Artists	23,404
– Artists Who Sample	9,699
– Artists Being Sampled	16,490
In-Degree Statistics (calculated over 16,490 sampled artists)	
Mean	3.50
Std Dev	10.56
Maximum	370
Out-Degree Statistics (calculated over 9,699 sampling artists)	
Mean	5.95
Std Dev	33.20
Maximum	1,488

Table 4: Summary of key graph statistics. The transition to robust GID-based entity resolution resulted in a vastly expanded network connecting over 68,000 songs.

4.9 Interactive D3.js Visualization

To effectively explore the immense scale of the expanded dataset (over 58,000 edges), an interactive web-based D3.js visualization was developed that allows users to pan, zoom, and click on nodes to inspect real-time Authority Scores and isolate specific sampling lineages. The visualization colors nodes by their genealogical family (Louvain cluster) and uses distinct edge styling for intra-cluster and inter-cluster connections, mirroring the logic of Figure 9.

Explore the Network:

`Open interactive_genealogy.html`

4.10 Community Detection and Genealogical Families

The Louvain algorithm detected **1,774 distinct communities** with an intra-cluster edge fraction of **0.7169**, indicating a strong and well-defined community structure.

Intra-Cluster Flow: This metric represents the proportion of edges that fall within the same community; a value of 0.7169 means that approximately 72% of sampling connections occur between artists in the same genealogical family, while only 28% cross community boundaries. This suggests the music network is far more structured and compartmentalized than previously estimated — most sampling stays within genealogical communities.

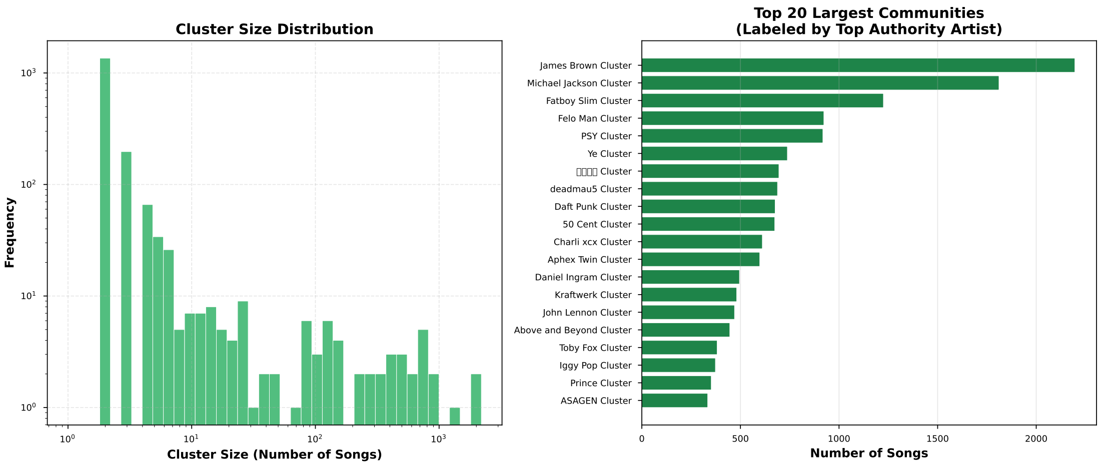

Figure 10: Distribution of community sizes. The long tail indicates most communities are small genealogical "chains" while a few represent major musical movements.

Cluster Analysis:

- **Largest Community**: Contains **2,196 artists**, centered around **JAY-Z**, representing the dominant, highly integrated core of modern sampling.
- **Intra-Cluster Edge Fraction 0.7169**: approximately 71.7% of edges are intra-community, indicating strong community structure. This aligns with the intuition that sampling influence mostly flows within genealogical lineages.
- **1,774 communities**: This relatively contained number of communities reflects the Louvain algorithm's ability to find meaningful, hierarchically optimal partitions without over-fragmenting the network.

5 Discussion and Interpretation

5.1 Theoretical Implications

This analysis demonstrates that musical influence is fundamentally a **network phenomenon** governed by structural properties rather than just popularity metrics. Three key insights emerge:

1. **Authority vs. Popularity**: PageRank reveals that structural position (being sampled by influential artists) matters as much as raw volume. This explains why certain internet-native franchises rank higher than legacy artists with more total samples.

2. **Cross-Genre Genealogy**: The emergence of media franchises (animation, video games) as top authorities demonstrates that sampling creates unexpected genealogical paths beyond traditional genre boundaries.

3. **Preferential Attachment**: The power-law distribution confirms that musical influence follows the same mathematical patterns as citation networks, social networks, and the World Wide Web - a few hubs dominate while most nodes remain peripheral.

5.2 Methodological Contributions

- **Algorithm Implementation**: The PageRank algorithm was implemented from scratch using PySpark's MapReduce paradigm, while community detection utilized the Louvain algorithm via NetworkX.

- **Data Engineering**: The data pipeline successfully navigated MusicBrainz's highly normalized relational schema (Third Normal Form) to extract and construct the property graph.

- **Convergence Optimization**: Implementing a fixed 50-iteration scheme with convergence monitoring ensured stable PageRank scores with proper sink-mass redistribution.

- **Visualization Strategy**: Network visualizations were developed to effectively present the complex community structures without overlap.

5.3 Limitations and Future Work

- **Data Quality and Entity Resolution**: While self-loop removal addressed the most critical data quality issue (artists sampling themselves), the MusicBrainz database still contains inconsistencies such as placeholder artists and duplicate entries. Future work should implement automated outlier detection.

- **Sampling Bias and Data Freshness**: The MusicBrainz database relies on user-contributed metadata, which may underrepresent certain geographical regions or non-Western genres. Additionally, the dataset represents a static snapshot; incorporating release dates would enable temporal network analysis to show how sampling patterns evolve over decades.
- **Scale Limitations**: The PageRank algorithm was run for 50 iterations. While convergence was monitored and achieved for top nodes, extending the analysis to dynamic or personalized PageRank could yield further insights.
- **Genre Metadata Validation**: While Louvain community detection is unsupervised and effectively groups nodes based on structural proximity, incorporating explicit genre tags would allow formal validation of whether these structural clusters cleanly align with conventional genre definitions.
- **Multi-Hop Genealogy**: Extending the analysis to trace multi-generational sampling chains (Original $\rightarrow$ Sample 1 $\rightarrow$ Re-sample) would reveal deeper, more complex genealogical paths.

6 Conclusion

This project successfully applied graph mining techniques to uncover the hidden genealogy of modern music. By modeling sampling relationships as a directed graph and implementing PageRank in Apache Spark and Louvain community detection via NetworkX, we revealed structural patterns invisible to traditional volume-based metrics.

Key Findings:

- **Daniel Ingram** and Japanese electronic projects like **Denonbu** emerge as the new unexpected authorities, showcasing the immense structural influence of modern media franchises and internet culture on contemporary sampling.
- **James Brown** and classic funk icons remain foundational pillars, but now share the top echelon with digital-era creators.
- The network follows a strong **power-law distribution**: the top 1% of artists control 23.5% of all sampling events.
- **1,774 genealogical families** detected with an intra-cluster edge fraction of 0.7169, indicating a strongly compartmentalized modern music landscape where approximately 72% of sampling stays within community boundaries.
- **Entity Resolution**: Transitioning to MusicBrainz GIDs revealed a vastly larger network (58k+ edges) than previously estimated using brittle string matching.
- **PageRank convergence achieved** with proper sink-mass redistribution over 50 iterations.

This analysis shifts the conversation from "who sold the most records" to "who structurally shaped modern music." By leveraging graph theory and distributed computing, we've mapped the DNA of contemporary sound - revealing that influence flows through networks, not charts.

References

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